HSV.AI

Approachable AI

Understanding AI in plain terms

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Introduction to J. Langley

Lots of titles:

- Chief Technical Officer
- Chief Archeologist
- Chief Instigator

I've been working with AI since around 2005, with my master's project at Florida Institute of Technology being an NLP based system for recommending web forum channels that best match a user's question.

> I believe that the best way to ensure that AI is used for the greater good is to involve the greatest number of perspectives possible in its development, testing, and use.

What We'll Cover Today

Part 1 — Understanding AI

- How AI language models came to be
- The scale of today's AI moment
- AI in your personal life — already, without knowing it
- AI already showing up in healthcare settings
- AI alignment — what it means & why it matters

Part 2 — Practical AI

- Levels of AI literacy — where do you fit?
- Why chat sessions can go off the rails
- AI for Health Unit Coordinators
- Building an AI policy for your organization

Goal: Walk away with a grounded, practical understanding of AI — and concrete steps to use it responsibly in your work.

How We Got Here: A Short History

1950s–1990s

Computers follow explicit rules — if a sentence matches a pattern, respond with a pre-written answer. Brittle and limited.

2013

Word2Vec: AI learns that words used in similar contexts carry similar meaning. Language starts becoming mathematical.

2017

Google publishes *"Attention Is All You Need"* — the **Transformer** breakthrough. Models can now understand context across long passages of text.

2018–2020

GPT, BERT — models trained on the entire internet begin to understand nuance, tone, and meaning the way humans do.

Nov 2022

ChatGPT launches publicly. 1 million users in 5 days. 100 million in 2 months. AI enters everyday life.

2023–2024

GPT-4, Claude, Gemini — models that can read images, write code, pass licensing exams, and reason through complex problems

2025–2026

AI **agents** — systems that can take actions, not just answer questions. AI moves from assistant to autonomous worker.

Today

AI is embedded in scheduling, electronic health records, and clinical tools — and your team is already using it.

The Scale of What's Happening

2 mo

For ChatGPT to reach 100M users — faster than any technology in history, including the smartphone

$500B+

Annual private investment in AI (2025) — more than the annual GDP of many countries

90%+

Of major health systems have at least one AI initiative underway (Deloitte Health, 2025)

> **A useful analogy:** Think of AI language models the way you think of the EHR transition. Not everyone needs to understand how the database works — but everyone needs to know how to use it safely, what it can get wrong, and what the rules are around patient data.

AI in Your Personal Life — Right Now, Without Realizing It

These tools feel normal. That's because AI is already woven into daily life — long before ChatGPT made the headlines.

Google Search
AI ranks every result, autocompletes your query, and filters spam

Spotify
Recommends songs and builds playlists based on your listening history

▶

Netflix / YouTube
Decides what appears on your home screen based on viewing patterns

Google Maps / Waze
Predicts traffic in real time and reroutes you before you hit slowdowns

Amazon
Recommends products and predicts what you'll want to reorder

Gmail / Email
Filters your spam, suggests replies, and completes sentences as you type

Face ID / Unlock
AI recognizes your face in milliseconds to unlock your phone

Social Media Feeds
AI decides which posts you see — and in what order — every time you scroll

> The AI in ChatGPT and Claude is built on the same underlying technology — just trained to have a conversation instead of rank a search result.

AI Is Also Showing Up in Healthcare

In Clinical Settings

- **Early warning & deterioration alerts** — some EHR platforms include AI-powered risk scores that run automatically in the background
  Examples: Epic Oracle Health
- **Radiology support tools** — AI can flag areas of concern on imaging for a radiologist to review
  Examples: Aidoc Nanox.AI
- **Drug interaction checking** — AI-enhanced drug databases built into pharmacy workflows that flag potentially dangerous combinations before dispensing
  Examples: FDB DrugBank
- **Ambient documentation** — tools that listen to a clinical encounter and draft a note
  Examples: Nuance DAX Suki

In Operations & Administration

- **Scheduling optimization** — AI tools that help manage staffing ratios and bed availability
  Examples: Qventus LeanTaaS
- **Prior authorization assistance** — tools that help draft and submit insurance requests
  Examples: Cohere Health Availity
- **Medical coding assistance** — AI that suggests billing codes from clinical notes
  Examples: 3M / Solventum Optum360
- **Readmission & risk prediction** — scores generated at or before discharge to flag high-risk patients
  Examples: Epic Oracle Health

Note: Adoption varies widely by health system. Your organization may use some, all, or none of these — the point is that the category of tools exists and is growing rapidly.

AI Alignment

Why "works most of the time" isn't good enough when real decisions are at stake

What Is AI Alignment?

Definition: AI alignment is the challenge of ensuring that an AI system does what the organization building or deploying it intends — including in edge cases, under pressure, and as systems become more capable. A system can technically meet its stated objective while causing real harm.

This is why involving the greatest number of perspectives in AI development, testing, and use matters — alignment to a single organization's goals may not serve the greater good unless those building and deploying AI actively seek broad input.

The Core Problem

- AI systems optimize for what they're *measured on* — not necessarily what patients need
- A model trained to reduce hospital returns might recommend downplaying risk factors
- Confident-sounding output is not the same as correct output
- More capable systems find more creative "shortcuts"

Clinical Analogy

Imagine a hospital alert system that fires correctly for medication issues — but so often that staff learn to click through without reading. Technically "working," but misaligned with patient safety.

Alert fatigue is an alignment problem. The system optimized for catching errors, not for being acted on.

Where Alignment Began: The HHH Framework

Early AI alignment work — including at Anthropic — converged on three foundational goals. The field has grown more nuanced since, but these concepts still shape how AI tools respond to you every day, with far-reaching implications for how you use them.

Helpful

Genuinely useful to the person asking — and, through that, to society. Not helpful in a watered-down, hedge-everything way, but substantively helpful the way a knowledgeable friend would be.

In healthcare: Give me a real answer about this medication, not just "consult your doctor."

✅ Honest

Truthful, transparent, and non-manipulative. Shares uncertainty when it exists. Never tries to create false impressions — even by technically true but misleading statements.

In practice: "I'm not certain — you should verify this before relying on it."

Harmless

Avoids outputs that could hurt individuals, groups, or society — even when asked. Maintains human oversight so errors can be caught and corrected before causing real damage.

In practice: Won't provide specific instructions that could cause harm without proper context.

The tension — and why it matters to you: These three goals pull against each other. Being *maximally helpful* could mean giving dangerous information. Being *maximally harmless* could make the tool useless. AI companies have continued to refine this balance, but the core trade-off remains — which is why human judgment and oversight still matter every time you use these tools.

The Alignment Spectrum

Now

Today's Real Risks

- AI generating wrong medical information — confidently
- Models trained on data that underrepresents minorities, women, older adults
- Patient information entered into consumer AI tools
- Overconfident outputs accepted without verification
- Widening health equity gaps via biased models

Emerging

Near-Future Challenges

- AI agents making critical patient care decisions without a human
- AI-generated insurance request denials with no human review
- Synthetic notes inserted into electronic medical records
- Liability when AI-assisted care causes harm
- Staff displacement without support structures

Long-Term

Broader Questions

- AI systems setting clinical standards without physician input
- Erosion of clinician judgment as AI dependency grows
- Concentration of healthcare AI power in few vendors
- Loss of human accountability in care decisions

The "Now" column is your organization's problem today. These are not hypothetical — they are happening in health systems without governance structures.

How AI Is Made Safer — and Why You Still Need Oversight

What AI Companies Do

- **Human feedback training (RLHF)** — clinicians, teachers, lawyers rate AI outputs to reinforce good behavior
- **Constitutional AI** — model evaluates its own responses against a set of principles before answering
- **Red-teaming** — adversarial testers deliberately try to make the model fail or produce harmful outputs
- **Usage policies** — explicit prohibited uses built into the model's training

Regulatory & External Checks

- **FDA oversight** — clinical AI tools that affect patient care are regulated as medical devices
- **NIST AI Risk Management Framework** — voluntary but widely adopted across healthcare
- **EU AI Act** — classifies healthcare AI as "high risk" with specific requirements
- **Federal transparency rules** — organizations must disclose when AI assists in clinical decisions

> **Think of it like a new drug:** FDA approval means it passed rigorous testing — not that it's right for every patient or that the prescriber can skip judgment.

Practical AI for Healthcare Professionals

Communications · Patient Flow · Scheduling · Documentation · Compliance · and more

Levels of AI Literacy

0 — **Unaware**
Uses AI daily without recognizing it — recommendation engines, autocomplete, automated alerts in the tools they already use. No mental model of how it works or where it can fail.

1 — **User**
Intentionally uses AI tools — ChatGPT or Claude to draft communications, summarize documents, or look up information. Understands that AI can be wrong and checks important outputs.

2 — **Practitioner**
Builds repeatable AI-assisted workflows — templates, structured prompts, evaluation criteria. Trains colleagues. Evaluates vendor AI tools critically.

3 — **Configurator / Analyst**
Configures or evaluates AI tools used by the organization, works with vendors and IT, designs governance processes, and analyzes whether AI is performing as expected.

4 — **Researcher / Builder**
Trains or fine-tunes AI models for specialized use. Contributes to AI research, bias detection, and alignment work. Often in academic or R&D roles.

Most healthcare staff today are Level 0–1. The organizational goal is Level 1 for all staff, Level 2 for department leads and coordinators.

Why Chat Sessions Can Go Off the Rails

What Is "Context"?

Every AI chat session has a **context window** — the total amount of text the model can actively "see" at once. Think of it like working memory: everything in the current conversation, including your messages and the AI's replies, lives in that window.

> **Hospital analogy:** Imagine giving a new resident a patient's entire chart to hold in their head — at some point, new information starts crowding out older details. The AI has the same constraint, just measured in tokens (chunks of text) instead of pages.

What Happens When It Fills Up

- The AI begins to lose track of **earlier parts of the conversation**
- Responses become **less coherent** or start contradicting prior answers
- The session may **error out** entirely once the limit is reached
- Mixing many different topics makes the problem worse — the AI has to juggle too much at once

How to Avoid It

One topic per session. Start a fresh conversation for each distinct task. Don't use the same chat for drafting a patient note, then asking a policy question, then requesting a template update — each one deserves its own session.

Keep prompts focused. Provide the specific information needed for that task — not the entire patient history unless it's all relevant. The less clutter in the context, the better the output.

When quality drops, start over. If the AI starts giving vague or contradictory answers, it's often a sign the context is getting crowded. Open a new chat and restate your request clearly.

> **Tip:** Name your chats descriptively ("Patient summary – West Wing" not "Chat 1") so you can find and reuse a focused session instead of piling unrelated tasks into one.

AI for Clinical Communications

Patient-Facing

- **Discharge instructions** — convert clinical notes into plain-language, literacy-appropriate instructions
- **After-visit summaries** — draft from encounter notes in seconds
- **Patient portal responses** — draft replies to patient messages for provider review and send
- **Education materials** — condition-specific content tailored to reading level and language
- **Informed consent summaries** — plain-language versions of complex procedures

Team & Administrative

- **Shift handoff notes** — structured summaries from nursing notes for incoming staff
- **Referral letters** — complete summaries from chart data
- **Meeting summaries & action items** — from recorded care team huddles
- **Policy drafts** — first versions of updated protocols from regulatory guidance

Real example: A nurse inputs key points from an encounter; Claude drafts a discharge instruction document with med list, follow-up steps, and warning signs — reviewed and edited in 2 minutes instead of 15.

AI for Health Unit Coordinators

Patient Flow & Admissions

- Drafting admission paperwork and consent summaries
- Generating transfer-of-care communication packets
- Discharge checklist creation from order sets
- Flagging incomplete orders before transfer
- Summarizing bed availability status updates

Scheduling & Coordination

- Drafting consult request messages from order notes
- Scheduling communication with radiology, lab, physical and occupational therapy
- Tracking and following up on pending orders
- Generating transport request summaries
- Pre-procedure prep instruction drafts for patients

Unit Communications

- Drafting family update messages from chart notes
- Creating shift bulletin or whiteboard content
- Translating clinical instructions to plain language
- Composing interdepartmental communications
- Summarizing lengthy policy updates for staff

Documentation & Records: Billing and coding assistance, incident report drafting, chart deficiency follow-up, forms completion guidance

Compliance: HIPAA policy lookup, Joint Commission prep, tracking regulatory updates, prior auth letter drafting

Supply & Logistics: Drafting supply order justifications, summarizing vendor communications, generating equipment request documentation

Your Organization Needs an AI Policy

Why Now?

- Staff *are already using* AI tools — with or without guidance
- Patient information is being typed into ChatGPT right now
- AI-generated clinical content is being used without clinician review
- HIPAA liability exposure is real when PHI enters consumer AI apps
- Regulatory requirements are arriving (ONC, state nursing boards)
- Staff need clarity on what's supported — and what's supported *by IT*

What Happens Without One

- A staff member pastes a patient's full history into ChatGPT to figure out what's wrong — HIPAA violation
- AI-drafted discharge instructions with wrong medication doses are sent to a patient
- A physician uses AI for a referral letter that cites a fabricated guideline
- Staff fear discipline for using AI, even when use is reasonable

This is already happening. Cybersecurity researchers found that 81% of healthcare workers' data policy violations involved regulated patient data — and 71% use personal AI accounts not covered by organizational data controls. (HIPAA Journal / Netskope, 2025)

AI Policy: What Mature Policies Address

1 **Approved Tools List**
Which AI tools are sanctioned by IT/Legal? Which require a Business Associate Agreement (BAA)?

2 **PHI / Data Rules**
Clear rules on patient identifiers in consumer AI tools, and a shared definition of what "de-identified" means for AI inputs.

3 **Clinical Oversight Requirements**
All AI-generated clinical content must be reviewed and attested to by a licensed clinician before use.

4 **Disclosure Standards**
When must patients, regulators, or accreditors be told that AI assisted in a clinical or administrative decision?

5 **Training Requirements**
Role-based AI literacy training; competency check for clinical AI tools; annual refresher as tools evolve.

6 **Incident Reporting**
Clear pathway to report AI errors, near-misses, or unexpected outputs — just like medication errors or equipment failures.

Start simple: A one-page policy covering PHI rules, approved tools, and clinical review requirements is far better than a perfect 50-page policy that takes 18 months to approve.

Key Takeaways

- **AI is already in your clinical environment.** The question is whether it's used intentionally and safely.
- **LLMs are reasoning assistants, not oracles.** They are fast, capable, and wrong often enough that human review matters.
- **Alignment is a patient safety issue.** Biased training data, incorrect information delivered confidently, and patient data leakage are real risks — not just IT concerns.
- **Literacy creates agency.** Staff who understand AI make better decisions about when to trust it — and when to override it.

How Organizations Typically Get Started

Week 1–2: Understanding what AI tools staff are already using — officially and unofficially — is usually the first step

Month 1: Most organizations start with a one-page interim policy covering PHI rules and approved tools, developed with Legal and Compliance

Month 2: Early pilots of 2–3 use cases (patient summaries, portal responses, handoff notes) help teams learn what works before scaling

Month 3: Organizations that sustain progress typically add role-based training, an AI champion at the department level, and a pathway for reporting AI-related errors

Questions?